西安电子科技大学 XIDIAN UNIVERSITY
物理与光电工程学院

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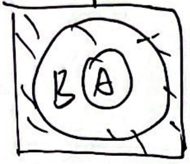
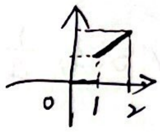
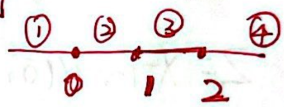
(2020级)
2021年秋期末考试（2022年12月12日）

一、1. 画图  $P(A|B)=0$



2. 计算 $F_Y(y)$，$f(x)=\begin{cases}\frac{1}{2}, & 0<x<2\\ 0, & \end{cases}$ $\boxed{\text{先确定Y的有效取值范围}}$ Y：0，[1,2]



$y<0$ 时，$F_Y(y)=0$

$0<y<1$ 时，$F_Y(y)=P(Y\le y)=P(Y=0)=P(X<1)=\frac{1}{2}$

$1<y<2$ 时，$F_Y(y)=P(Y\le y)=P(X\le y)=\int_{-\infty}^{y}\frac{1}{2}dx=\int_0^y\frac{1}{2}dx=\frac{1}{2}y$

$y>2$ 时，$F_Y(y)=1$

$\therefore$ ~~$F_Y(y)=\begin{cases}0, & y<0\\ 0<y<1\text{时}, & \end{cases}$~~ $F_Y(y)=\begin{cases}0, & y<0\\ \frac{1}{2}, & 0\le y<1\\ \frac{1}{2}y, & 1\le y<2\\ 1, & y\ge 2\end{cases}$ 0一个间断点

3. 独立。$f_X(x)=\begin{cases}e^{-x}, & x>0\\ 0, & x\le 0\end{cases}$ $f_Y(y)=\begin{cases}4e^{-4y}, & y>0\\ 0, & y\le 0\end{cases}$ $f(x,y)=f_X\cdot f_Y=4e^{-x-4y}$

$P(X<Y)=\iint_{x<y}f(x,y)dxdy=\iint_{x<y}4e^{-x-4y}dxdy$

$=\int_0^{+\infty}dx\int_x^{+\infty}4e^{-x-4y}dy=\int_0^{+\infty}e^{-5x}dx=-\frac{1}{5}(e^{-5x}\Big|_0^{+\infty})=-\frac{1}{5}(0-1)=\frac{1}{5}$

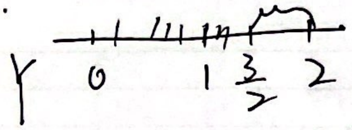
4. 辛钦大数定律：独立，同分布，~~期望存在且有界~~ $f(x)=F'(x)=$

$\int|x|f(x)dx=+\infty$ 不适用

5. $X_1-X_2\sim N(0,2\sigma^2)$，$\frac{X_1-X_2}{\sqrt{2}\sigma}\sim N(0,1)$；$X_3\sim N(0,\sigma^2)$，$\frac{X_3}{\sigma}\sim N(0,1)$

$(\frac{X_3}{\sigma})^2\sim\chi^2(1)$ $\therefore\ \frac{\frac{X_1-X_2}{\sqrt{2}\sigma}}{\sqrt{(\frac{X_3}{\sigma})^2}}=\frac{X_1-X_2}{\sqrt{2}|X_3|}\sim t(1)$

二. 1.



$P(Y\le\frac{3}{2}) = 1-P(\frac{3}{2}<Y\le 2) = 1-\frac{1}{2}\times\frac{1}{4} = \frac{7}{8}$

$0\le Y\le 2$

2. $P(X\le\frac{1}{2}) = \int_0^{\frac{1}{2}} 2x\,dx = x^2\Big|_0^{\frac{1}{2}} = \frac{1}{4}$

$Y\sim B(3,\frac{1}{4})$, $P(Y=2) = C_3^2(\frac{1}{4})^2(\frac{3}{4})^1 = \frac{9}{64}$

3. $\rho=0$ 故 X,Y 相互独立. 故 $X\sim N(1,1)$, $Y\sim N(0,1)$

$Z=X-1\sim N(0,1)$, $Y\sim N(0,1)$

互斥

$P(XY-Y<0) = P((X-1)Y<0) = P(ZY<0) = P(Z<0,Y>0 \cup Z>0,Y<0)$

$= P(\underbrace{Z<0,Y>0}_{\text{独立}}) + P(\underbrace{Z>0,Y<0}_{\text{独立}})$

$= \frac{1}{2}\times\frac{1}{2}+\frac{1}{2}\times\frac{1}{2} = \frac{1}{2}$

4. $EU_i = E(X_i+X_{n+i}-2\bar{X})^2$ 设 $X_i+X_{n+i}-2\bar{X}=Y$

$= EY^2 = DY+EY^2$ $X_i, X_{n+i}, \bar{X}$ 不独立.

$EY = E(X_i+X_{n+i}-2\bar{X}) = 0$

$DY = D[X_i+X_{n+i}-\frac{1}{n}(X_1+\cdots X_i+\cdots+X_{n+i}+\cdots+X_{2n})]$

$= D[\frac{n-1}{n}(X_i+X_{n+i}) - \frac{1}{n}\underbrace{(X_1+\cdots X_{i-1}+X_{i+1}\cdots+X_{n+i-1}+X_{n+i+1}+\cdots X_{2n})}_{2n-2\text{项}}]$

$= \frac{2(n-1)^2}{n^2}\cdot\sigma^2 + \frac{1}{n^2}(2n-2)\sigma^2$

$= \frac{2(n-1)\sigma^2}{n}$

5. $\boxed{T=\frac{\bar{X}-\mu}{s/\sqrt{n}}}$, ~~$P(|\frac{\bar{X}-\mu}{s/\sqrt{n}}|<t_{\frac{\alpha}{2}}(8)) = 1-\alpha = 0.95$~~

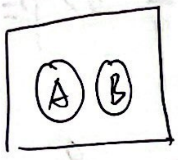
找对称轴极轴号，边界点.

$P(a<\frac{\bar{X}-\mu}{s/\sqrt{n}}<b) = P(-t_{\frac{\alpha}{2}}<\frac{\bar{X}-\mu}{s/\sqrt{n}}<t_{\frac{\alpha}{2}}) = 0.95$

$\bar{x}=6$, $s=0.5$, $n=9$, $t_{\frac{\alpha}{2}}(8) = t_{0.025}(8) = 2.306$

$-2.306<|\frac{6-\mu}{0.5/3}|<2.306 \Rightarrow \mu\in(5.616, 6.384)$

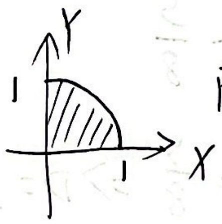
2020年期末(2019级)

一. 1.  画图+排除. $P(\bar{A}\cup\bar{B})=1$

2. 相互独立定义: $F(x,y)$, 若 $F_X(x)$, $F_Y(y)$, 如 $F(x,y)=F_X(x)F_Y(y)$ 则 X,Y 相互独立.

定理1: 二维离散型, X,Y 相互独立 $\Leftrightarrow p_{ij}=p_{i\cdot}p_{\cdot j}$

定理2: 二维连续型: $f(x,y)$, $f_X(x)$, $f_Y(y)$ 是除有限个点外连续函数, 则 X,Y 相互独立 $\Leftrightarrow f(x,y)=f_X(x)f_Y(y)$

3.  面积为 $\frac{\pi}{4}$, $f_X(x)=1$, $f_Y(y)=1$, X,Y 相互独立, $f(x,y)=\begin{cases}1, & 0<x<1,\ 0<y<1\\ 0, & 其他\end{cases}$

故 $P(X^2+Y^2\le 1)=\iint_{x^2+y^2\le 1} 1\,dxdy=\frac{\pi}{4}$

4. $\sum_{i=1}^{100}X_i \sim N(n\mu, n\sigma^2)=N(100\times\frac{1}{2}, 100\times\frac{1}{4})$

$B(1,p)$: $EX=p$, $DX=pq$

$$P\left(\sum_{i=1}^{100}X_i\le 55\right)=P\left(\frac{\sum_{i=1}^{100}X_i-50}{5}\le\frac{55-50}{5}\right)=P\left(\frac{\sum_{i=1}^{100}X_i-50}{5}\le 1\right)=\Phi(1)$$

5. $X\sim t(n)$ $(n>1)$ 则 $X=\frac{X_1}{\sqrt{X_2/n}}$ 其中 $X_1\sim N(0,1)$, $X_2\sim\chi^2(n)$

$X^2=\frac{X_1^2}{X_2/n}=\frac{nX_1^2}{X_2}$, 则 $X_1^2\sim\chi^2(1)$, $X_2\sim\chi^2(n)$

$\therefore \frac{1}{X^2}\sim F(n,1)$

二. 1. $P(A|B)=\frac{P(AB)}{P(B)}=\frac{P(AB)}{0.8}$, 需计算 $P(AB)$

~~P(~~ 已知 $P(B|\bar{A})=0.6=\frac{P(\bar{A}B)}{P(\bar{A})}=\frac{P(\bar{A}B)}{0.4}\Rightarrow P(\bar{A}B)=0.24$

$P(\bar{A}B)=P(B)-P(AB)=0.8-P(AB)=0.24\Rightarrow P(AB)=0.56$

$\therefore P(A|B)=\frac{0.56}{0.8}=\frac{7}{10}=0.7$

2. $P(X\geq 1) = \cancel{1-P(P<}\ 1-P(X<1) = 1-P(X=0) = 1-C_2^0(p)^0(1-p)^2 = \frac{5}{9} \Rightarrow p=\frac{1}{3}$

$\therefore Y \sim B(3,\frac{1}{3})$

$P(Y\geq 1) = 1-P(Y<1) = 1-P(Y=0) = 1-C_3^0(\frac{1}{3})^0(\frac{2}{3})^3 = 1-\frac{8}{27} = \frac{19}{27}$

3. $EX=EY=\frac{3}{4}$, $DX=DY=\frac{3}{16}$

$D(X-Y)=\frac{1}{4} = DX+DY-2Cov(X,Y) \Rightarrow Cov(X,Y)=\frac{1}{16}$

$Cov(X,Y) = EXY - EXEY = EXY-\frac{9}{16} = \frac{1}{16} \Rightarrow EXY=\frac{5}{8}$

Y\X	0	1	
0	A	B	$\frac{1}{4}$
1	C	D	$\frac{3}{4}$
	$\frac{1}{4}$	$\frac{3}{4}$	

XY	0	1
	1-D	D

$EXY = 1\times D = \frac{5}{8} \Rightarrow D=\frac{5}{8}$

可得 $A=B=C=\frac{1}{8}$

Y\X	0	1
0	$\frac{1}{8}$	$\frac{1}{8}$
1	$\frac{1}{8}$	$\frac{5}{8}$

$P(X+Y\leq 1) = 1-P(X+Y>1)$
$= 1-P(X+Y=2)$
$= 1-P(X=1, Y=1)$
$= 1-\frac{5}{8} = \frac{3}{8}$

4. $(\bar{X}\pm\delta)$, $\bar{x}=9.5$, $\bar{x}+\delta=10.8$, $\delta=1.3$ 则下限为 $\bar{x}-\delta=8.2$

5. 接受双边检验，则

$\left|\frac{\bar{x}-\mu}{\sigma/\sqrt{n}}\right| < z_{0.025}=1.69$

$\mu=30$, $\sigma=4$, $n=100$ 代入

$\bar{x}\in(29.216, 30.784)$